

Baseline Findings (Before Quantum)

Single Source of Truth

Qiskit Fall Fest 2026, Use Case 05: Aviation emissions and flight-path optimisation.
Prepared for the team on 8 October 2026.

1. Purpose of this document

This document records every "before" number for our project. After we run the quantum optimisation, we compare the new results only against the numbers written here. If a number changes, we change it here first and tell the whole team.

2. Data source

We use the file `flights_clean.csv`. It holds completed flights for November 2025 (1 to 25 November) that touch California airports. Each row is one flight and includes delays, taxi times, aircraft type, and the emissions of the landing and take-off cycle.

Item	Value
Flights in the file	84,619
Flights with emissions values	81,585
Flights without emissions values	3,034
Origin airports	99
Different aircraft (tail numbers)	4,219
Busiest origin airport	LAX, with 12,295 departures

One warning for the team: the data dictionary says there are 103,851 completed flights, but `flights_clean.csv` has 84,619 rows. We use the 84,619 rows because that is the file we actually analyse. We should be ready to explain this difference if a judge asks.

3. What the emissions numbers cover

The emissions in the file cover only the ICAO landing and take-off (LTO) cycle. That means take-off, climb-out to 3,000 feet, approach, and taxiing. Cruise emissions are not in the data, so we must always say "LTO emissions" and never "total flight emissions" in our slides.

4. Baseline for the whole dataset

Measure	Before value
Total LTO CO2	214,796 tonnes
Average LTO CO2 per flight	2,633 kg
Average LTO fuel per flight	833 kg
Total taxi time (taxi-out plus taxi-in)	2,312,361 minutes
Average taxi-out time	19.21 minutes
Average taxi-in time	8.12 minutes

Measure	Before value
Delay caused by a late aircraft	403,118 minutes
Delay-chain flights (follow an earlier flight of the same aircraft)	62,465
Estimated CO2 from taxiing	about 99,711 tonnes

The taxi CO2 is an estimate, not a measured value. See the assumptions in section 7.

5. Congestion finding at LAX

We measured how taxi-out time changes when more flights leave the gate in the same 15-minute window at LAX. The fitted relation is:

Taxi-out time = 16.83 minutes + 0.310 minutes for every extra departure in the same 15-minute window.

The correlation is weak ($r = 0.11$). This means the effect is real but small, and it limits how much any optimiser can save. We must state this honestly in the presentation.

6. The test case we will optimise

Both the classical and the quantum methods must work on exactly the same test case.

Item	Value
Airport	LAX
Day	7 November 2025
Time window	05:00 to 12:00 (actual gate departure)
Flights in the test case	203 (only flights with emissions values)
Estimated taxi CO2 of these flights	about 277,579 kg
LTO CO2 of these flights	about 589,490 kg
Busiest 15-minute window	14 departures

The decision for each flight is how long it waits at the gate with engines off: 0, 5, 10, or 15 minutes. The goal is to spread out departures so fewer planes sit in a queue with engines running.

Because real IBM quantum hardware can only handle a small number of qubits, the real-hardware run will use a much smaller piece of this test case, for example 4 to 6 flights from the busiest window. The baseline for that piece must be calculated and written in section 9 before the run.

7. Assumptions behind the numbers

These assumptions must stay the same in the "after" calculation.

1. CO2 is calculated as 3.16 kg of CO2 for every 1 kg of jet fuel.
2. We assume taxiing uses about 40 percent of the LTO fuel, and the ICAO standard taxi time of 26 minutes is used to turn that into fuel per taxi minute. This 40 percent is our assumption and must be checked against the ICAO Engine Emissions Databank.

3. Taxi time changes with congestion by 0.310 minutes for each extra departure in the same 15-minute window, as fitted in section 5.
4. Holding a flight at the gate adds no fuel burn because engines are off. The cost of waiting is counted as delay minutes.
5. The weight for one minute of gate waiting is 0.20 (in taxi-minute units), and the rule that each flight gets exactly one waiting choice is enforced with a penalty of 40.

8. Classical reference results (not quantum)

We already ran two classical methods on the test case in section 6. These are not the baseline and not quantum results. They are a fair yardstick for the quantum method. Please note that no real quantum hardware result exists yet.

Measure	Classical greedy	Simulated annealing (classical)
CO2 saved	2,797 kg	2,154 kg
Taxi minutes saved	62	48
Gate waiting minutes added	575	1,155
Flights held at the gate	80	129
Busiest window after	11 departures	12 departures

The saving is about 1 percent of the estimated taxi CO2 in the window. We should present this as a proof of method that can grow, not as a large saving.

9. Comparison table to fill after the quantum run

Only the "Before" column is fixed today. The other columns are filled after the run, using the same assumptions.

Measure	Before	Classical result	Real IBM hardware result
CO2 (kg) for the chosen small piece	Write here before the run	To fill	To fill
Taxi minutes saved	0	To fill	To fill
Gate waiting minutes added	0	To fill	To fill
Busiest window (departures)	Write here before the run	To fill	To fill
IBM machine name, job ID, shots	Not applicable	Not applicable	To fill

10. Rules for the after comparison

1. Use the same data file, the same test case, and the same assumptions as in this document.

2. Calculate the "after" CO2 with the same formula, so the difference comes only from the optimisation.
3. Throw away any quantum answer that breaks the rule of one waiting choice per flight, and report how many answers were thrown away.
4. Report the result honestly, even if the quantum answer only matches or is worse than the classical one.
5. Do not change any number in sections 2 to 8 without updating this document first.

11. Not yet covered

Delay chains are measured in the baseline (62,465 chain flights and 403,118 late-aircraft minutes), but our current optimisation does not change them. If we add them to the optimisation later, we must add a new row to section 9 and keep these baseline values.